Product Name: Nufarm Expand Adjuvant

APVMA Approval No.: 90892/136341

RLP
APPROVED

Label Name:	Nufarm Expand Adjuvant

Signal Headings:	READ SAFETY DIRECTIONS BEFORE OPENING OR USING

Constituent Statements:	ACTIVE CONSTITUENT: 500 g/L TRISILOXANE ETHOXYLATE

Mode of Action:	

Statement of Claims:	For use with low volume application to enhance the spreading and foliage deposition of fungicides, insecticides, plant growth regulators and foliar nutrients as per Directions for Use.

Net Contents:	5 L - 1000 L

Restraints:	DO NOT use with products where the label prohibits use of surfactants. DO NOT use Nufarm Expand if heavy rain or overhead irrigation is expected. DO NOT use in dilute spraying volumes as excessive run-off may occur.

Directions for Use:	This section contains file attachment.

Other Limitations:	

Withholding Periods:	WITHHOLDING PERIOD: REFER TO THE APPROPRIATE WITHHOLDING PERIOD OF THE AGRICULTURAL CHEMICAL THAT IS TO BE MIXED WITH NUFARM EXPAND.

Trade Advice:	

General Instructions:	Nufarm Expand is a non-ionic organosilicone surfactant blend designed specifically for the application of fungicides, insecticides, plant growth regulators and foliar nutrients in horticultural and arable crops. Nufarm Expand lowers the surface tension of spray solutions well below commonly used surfactants, but, unlike most other silicone 'super-wetters', Nufarm Expand will not give unwanted stomatal infiltration when used at recommended rates. Lower surface tension will result in more effective wetting and more uniform coverage on nearly any surface. Nufarm Expand will also enhance the amount of product retained on the leaf surface at spraying and will enable users to greatly reduce application volumes in most situations. Nufarm Expand readily disperses in spray solutions (aqueous and most non-aqueous). Nufarm Expand is especially suited to situations where coverage is critical and is particularly suited to low volume spraying where water coverage is a limiting factor. Use Nufarm Expand if the forecast for the next four days predicts dry weather or light showers. If heavy rain or overhead irrigation occurs after Nufarm Expand application, it may lead to pesticide loss due to runoff. In this case substitute Nufarm Expand for Designer® if heavier rain or overhead irrigation is likely to occur and enhanced rainfast properties are required. MIXING 1. Fill spray tank half full with water and begin agitation. 2. If foaming is expected, add an antifoaming agent to the tank before adding the pesticides/foliar nutrients. 3. Add pesticides/foliar nutrients as directed on their respective labels. Maintain agitation. 4. Shake Nufarm Expand container thoroughly and add the required amount after all the pesticides/foliar nutrients are thoroughly mixed in the tank. APPLICATION Adjust rates of Nufarm Expand and/or water volumes to obtain thorough coverage. Apply in spray volumes of 20-30% of the diluted spray volume (5x-3x). Optimum spreading and wetting from Nufarm Expand will occur between pH5 and pH9. Spray mixtures above and below these figures should be sprayed immediately or adjust the solution to within the correct range to avoid breakdown of the Nufarm Expand. COMPATIBILITY Nufarm Expand is compatible with most commonly used fungicides, insecticides, plant growth regulators and foliar nutrients provided that these products are suited to low volume application. Nufarm Expand should not be used with herbicides or products where the label prohibits use of surfactants. Nufarm Expand may be used with plant growth regulators in Horticultural tree crops provided the PGR is suitable for use in reduced spray volumes. Growers should check the performance of Nufarm Expand with a PGR on a small scale before wider use. Nufarm Expand may be used with particle film sunscreens e.g. Surround*. DO NOT mix Nufarm Expand with spray oils as phytotoxicity may occur. Nufarm Expand will break down rapidly when mixed with highly acidic or highly alkaline chemicals and poor wetting may result. Consult your local Nufarm representative for up to date information on product compatibility and spray application techniques. CAUTION Nufarm Expand has demonstrated good crop safety, however, not all plants and spray mixtures have been tested. Treatments that have a potential for phytotoxicity could be aggravated by Nufarm Expand. Before spraying any previously untried mixtures with Nufarm Expand, always test the mixture for crop safety. Before treating large areas, a small area should be tested for crop safety.

Resistance Warning:	

Precautions:	RE-ENTRY PERIOD DO NOT allow entry into treated areas until the spray has dried, unless wearing cotton overalls buttoned to the neck and wrist (or equivalent clothing) and chemical resistant gloves. Clothing must be laundered after each day's use.

Protections:	PROTECTION OF WILDLIFE, FISH, CRUSTACEANS AND ENVIRONMENT Dangerous to fish. DO NOT contaminate streams, rivers or waterways with the chemical or used containers.

Storage and Disposal:	KEEP OUT OF REACH OF CHILDREN. Store in the closed, original container in a cool, well ventilated area. DO NOT store for prolonged periods in direct sunlight. DO NOT store below 4ºC as product will freeze. DO NOT reuse empty container. Triple-rinse containers before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush, or puncture and deliver empty packaging to an approved waste management facility. If an approved waste management facility is not available, bury the empty packaging 500 mm below the surface in a disposal pit specifically marked and set up for this purpose, clear of waterways, desirable vegetation and tree roots, in compliance with relevant local, state or territory government regulations. Do not burn empty containers. Refillable containers Store in the closed original container in a cool, well-ventilated area. DO NOT store for prolonged periods in direct sunlight. Empty contents fully into application equipment. Close all valves and return to point of supply for refill or storage.

Safety Directions:	Will irritate the skin. Will damage the eyes. Avoid contact with eyes and skin. When opening the container, mixing and loading and preparing spray, wear cotton overalls buttoned to the neck and wrist (or equivalent clothing) and elbow-length chemical resistant gloves and goggles. When using the prepared spray, wear cotton overalls buttoned to the neck and wrist (or equivalent clothing) and elbow length chemical resistant gloves. If product on skin, immediately wash area with soap and water. If product in eyes, wash it out immediately with water. Follow all safety directions and precautions for the products with which Nufarm Expand will be mixed. Wash hands after use. After each day's use, wash gloves, goggles and contaminated clothing.

First Aid Instructions:	If poisoning occurs, contact a doctor or Poisons Information Centre. Phone Australia 13 11 26, New Zealand 0800 764 766.

First Aid Warnings:	

DIRECTIONS FOR USE

SITUATION	CROPS	RATE	CRITICAL COMMENTS
Use only with Fungicides, Insecticides, Plant Growth Regulators, Foliar Nutrients	**Vines**		NOTE: Unlike many other adjuvants, Nufarm Expand is applied at a per hectare rate not at a per 100 L water rate. Adjust rates of Nufarm Expand and/or water volumes to obtain thorough coverage. As Nufarm Expand is affected by many product formulations, always refer to product label compatibility section for more information. Use higher rate for lower water volumes, or when more than two pesticides are included in the tank mix. Increase water volume as canopy density increases. The performance of some products may be affected by application in low volume with Nufarm Expand, refer to the Compatibility section of this label. Nufarm Expand may be used with plant growth regulators in Horticultural tree crops provided the PGR is suitable for use in reduced spray volumes. Growers should check the performance of Nufarm Expand with a PGR on a small scale before wider use. **Spray Volume** Apply in spray volumes of 20-30% of the dilute spray volume (5x-3x). **Vegetables** DO NOT use in water volumes greater than 250 L/ha.
	Foliage	150-300 mL/ha	
	Bunches	400-600 mL/ha	
	Vegetables		
	Hard to wet e.g. onion	150-600 mL/ha	
	Easy to wet e.g. potatoes	100-400 mL/ha	
Use only with Fungicides, Insecticides, Plant Growth Regulators, Foliar Nutrients	**All horticultural tree crops including but not limited to:** mangoes avocados apples citrus olives	150-600 mL/ha	

NOT TO BE USED FOR ANY PURPOSE OR IN ANY MANNER CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION